



Dr. Kaushal Purohit

Home : Ratnam Garnet, Bhat Motera link road, behind Akshar Kutir, GIDC Bhat, Bhat, 382424, Ahmedabad, Gujarat, India

Email: kaushalsep1996@gmail.com **Phone:** (+91) 9427712691

Phone: (+91) 9316234342

Gender: Female **Date of birth:** 06/08/1996 **Nationality:** Indian

ABOUT ME

I am a hardworking, talented student who loves making a difference in the work. I am open-minded, patient, and supportive towards work.

WORK EXPERIENCE

[2018 – 2019]

Assistant lecturer

Sir P.T Science College <https://sirptsciencecollege.org>

Address: SCIENCE COLLEGE, College Campus, Dhansura Road, Modasa, India

Email address: sirptscience127@gmail.com

Physics Faculty

EDUCATION AND TRAINING

[2010 – 2011]

Secondary School Examination

K.N. Shah Modasa High School

Address: Station Road, At.Po.Ta.Modasa,Di.Sabarkatha, Modasa, 383315, Gujarat, India

Final grade: 78%

[2011 – 2013]

Senior Secondary Certificate Examination

K.N. Shah Modasa High School

Address: Station Road, At.Po.Ta.Modasa,Di.Sabarkatha, Modasa, 383315, Gujarat, India

Final grade: 60%

[2013 – 2016]

Bachelor of Science in Physics

Sir P.T Science College <https://sirptsciencecollege.org>

Address: Science College, College Campus, Dhansura Road, Modasa, 383315, Gujarat, India

Final grade: 9.0 CGPA

Thesis: Physics

[2016 – 2018]

Master of Science in Physics

Sardar Patel University <https://www.spuvvn.edu>

Address: Saheed chowk, Nana bazaar, Vallabh Vidhyanagar, Anand, 388120, Gujarat, India

Final grade: 6.51 CGPA

Thesis: Physics

[06/2019 – 09/2023]

Ph.D. in Physics

Sardar Vallabhbhai National Institute of Technology <https://www.svnit.ac.in>

Address: Ichchhanath Surat- Dumas, Road, Keval Chowk, Surat, 395007, Gujarat, India

Thesis: Eigensolution of the various potential and its application in different fields

PUBLICATIONS

[2020]

Eigensolution and various properties of the screened cosine Kratzer potential in D dimensions via relativistic and non-relativistic treatment

Reference: Kaushal R. Purohit, Rajendrasinh H. Parmar & Ajay Kumar Rai

The European Physical Journal Plus, Volume 135, Article number: 286 (2020)

<https://doi.org/10.1140/epjp/s13360-020-00299-7>

[2021]

Bound state solution and thermodynamic properties of the screened cosine Kratzer potential under influence of the magnetic field and Aharanov-Bohm flux field

Reference: Kaushal R. Purohit, Rajendrasinh H. Parmar & Ajay Kumar Rai

Annals of Physics, Volume 424 January 2021, 168335

<https://doi.org/10.1016/j.aop.2020.168335>

[2021]

Energy and momentum eigenspectrum of the Hulthén-screened cosine Kratzer potential using proper quantization rule and SUSYQM method

Reference: Kaushal R. Purohit, Rajendrasinh H. Parmar & Ajay Kumar Rai

Journal of Molecular Modeling, volume 27, Article number: 358 (2021)

<https://doi.org/10.1007/s00894-021-04965-0>

[2022] **Quarkonium spectroscopy of the linear plus modified Yukawa potential**

Reference: Kaushal R Purohit, Pooja Jakhad and Ajay Kumar Rai

Physica Scripta, Volume 97, Number 4

<https://doi.org/10.1088/1402-4896/ac5bc2>

[2022]

Solution of the modified yukawa-kratzer potential under influence of the external field and its thermodynamic properties

Reference: Kaushal R. Purohit, Ajay Kumar Rai & Rajendrasinh H. Parmar

Journal of Mathematical Chemistry, 60(9)

<https://doi.org/10.1007/s10910-022-01397-w>

[2023]

Spectroscopy of Heavy-Light mesons (cs,cq,bs,bq) for the linear plus modified yukawa potentials using NU method

Reference: Kaushal R. Purohit, Ajay Kumar Rai & Rajendrasinh H. Parmar

<https://doi.org/10.1007/s12648-023-02852-3>

[2023]

Momentum eigenspectrum of the modified Yukawa-Kratzer potential model via SUSYQM method

Reference: Kaushal R. Purohit, Ajay Kumar Rai & Rajendrasinh H. Parmar

Molecular Physics

Accepted

[2023]

Solution of Schrodinger equation in hot and dense QCD media using NU and SUSYQM method

Reference: Kaushal R. Purohit, Ajay Kumar Rai & Rajendrasinh H. Parmar

Under submission

[2023]

Study of few diatomic molecules under modified kratzer potential plus modified quantum interaction potential in D-dimensions

Reference: Kaushal R. Purohit, Ajay Kumar Rai & Rajendrasinh H. Parmar

Under writing

CONFERENCES AND SEMINARS

[2019]

Eigensolution of the modified screened cornell potential and mass spectroscopy of the charomonium

Kaushal Purohit, Ajay kumar Rai, & Rajendrasinh.H.Parmar

Proceeding of the DAE symp. on Nuclear Physics, 64,661

[2020]

Quarkonium spectroscopy via linear plus inverse square yukawa potential model

National Institute of Science Education and Research, Jatni, Odisha

Kaushal Purohit, Ajay kumar Rai, & Rajendrasinh.H.Parmar

XXIV DAE-BRNS Symposium on high Energy Physics

[2020]

Rotational Vibrational partion function using attractive radial potential plus class of yukawa potential

Govt. Engineering college, Bikaner, India

Kaushal Purohit, Ajay kumar Rai, & Rajendrasinh.H.Parmar

International Conference on Condensed Matter & Applied Physics.

American Institute of Physics Conference proceedings, 2020(1), 120004

[2020]

Approximate analytical solution of the extented Hulthen-Yukawa with inverse square and coulombic term plus ring shape potential

Govt. Engineering college, Bikaner, India

Kaushal Purohit, Ajay kumar Rai, & Rajendrasinh.H.Parmar

International Conference on Condensed Matter & Applied Physics.
American Institute of Physics Conference proceedings, 2020(1), 140071

[2021]

Mass spectrum of heavy quarkonium for harmonic plus screened Kratzer potential using series expansion method

Kaushal Purohit, Ajay kumar Rai, & Rajendrasinh.H.Parmar

Proceeding of the DAE Symp. on Nuclear Physics,65,533

[2022]

Bound state solution of the Schrodinger equation for attractive radial potential plus class of Yukawa potential and its statistical mechanical properties

Indian Institute of Teacher Education, Gandhinagar

Kaushal Purohit, Ajay kumar Rai, & Rajendrasinh.H.Parmar

National Conference on Advances in Physics Sciences for Sustainable Development

[2022]

Solution of the harmonic plus modified yukawa-kratzer potential and its application in hadron physics

Cotton University, Guwahati

Kaushal Purohit, Ajay kumar Rai, & Rajendrasinh.H.Parmar

66th DAE Symposium on Nuclear Physics

[2022]

Solution of the harmonic plus modified yukawa-kratzer potential and mass spectra of hidden charm and hidden strange tetraquark

Indian Institute of Science Education and Research, Mohali

Kaushal Purohit, Ajay kumar Rai, & Rajendrasinh.H.Parmar

XXV DAE-BRNS High Energy Physics Symposiums

WORKSHOPS

[2021] **Annual Physics workshop on " Advances in applied Physics" (AAP-2021)**

Department of Physics, SVNIT

[2021] **National Conference on Plasma Science and Applications (PSA-2021)**

Department of Physics, SVNIT

[2022] **Workshop on Nuclear Physics (NPW-2022)**

Department of Physics, SVNIT

LANGUAGE AND DIGITAL SKILLS

Languages Known

Gujarati, Hindi, English

Digital Skills

Matematica, Latex, Matlab, Phyton, Microsoft Office etc.

BOOKS

Application of Potential in Physical Sciences

Cambridge Research Scholar (Under publication)
